

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

SCORE SAVED TO COURSE PROGRESS

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1

Write pseudocode for a linear search and state its worst-case time complexity.

[5 marks]

 $O(n)$ *Method sound; final simplification omitted.*

5/5

2

Explain the difference between a stack and a queue, giving one appropriate use for each.

[5 marks]

Check the condition before applying the rule.

5/5

AGENT ANNOTATED

3

Trace the algorithm on the input [5, 2, 9, 1] and state the array after each pass.

[6 marks]

Correct and clearly justified.

5/6

4

Convert the binary number to denary and explain the place-value method used.

[4 marks]

 $1011\ 0110_2$

Method sound; final simplification omitted.

3/4

5

Describe how a hash table resolves collisions, naming one strategy and its drawback.

[6 marks]

Check the condition before applying the rule.

6/6

AGENT ANNOTATED

6

Write a function that returns the maximum value in a list without using a built-in max.

[5 marks]

Correct and clearly justified.

4/5

7

Explain why binary search requires a sorted list, and state its complexity.

[5 marks]

 $O(\log n)$

Method sound; final simplification omitted.

5/5

8

Compare compiled and interpreted execution, giving one advantage of each.

[5 marks]

Check the condition before applying the rule.

4/5